

AI-Driven Crime Trend Analysis in India: A Multi-Model Comparative Study

1st Dheeraj Kumar Mishra

Electrical Engineering

Indian Institute of Technology Bombay

Mumbai, India

25m1066@iitb.ac.in

2nd Dhanushk Sonkusare

Computer Science and Engineering

Indian Institute of Technology Bombay

Mumbai, India

25m0751@iitb.ac.in

Abstract—We present a comparative machine-learning study of India’s criminal-justice investigation pipeline using four consecutive years (2018–2021) of the National Crime Records Bureau (NCRB) Crime in India reports. After harmonizing schemas across the four publications and engineering a leakage-free feature set, we construct a master dataset of 519 crime-head-year observations spanning 125 unique Indian Penal Code (IPC) crime heads. We predict two operationally meaningful pipeline outcomes—the chargesheeting rate and the pendency percentage—using seven models (two naive baselines, linear regression, Ridge, Lasso, Random Forest, Support Vector Regression, and Gradient Boosting) under time-series cross-validation. A per-crime historical-mean baseline achieves RMSE = 8.74 on chargesheeting rate and is *not* outperformed by any supervised learner, indicating that between-crime differences dominate year-to-year variance. For pendency percentage, Lasso narrowly wins at RMSE = 8.03 with tree ensembles close behind, and the Random Forest assigns 89% of its importance to a single engineered feature, `pendency_load`. An unsupervised K-Means ($k = 3$, silhouette = 0.459) recovers three interpretable crime-behaviour profiles: a large well-prosecuted group, a property-crime backlog group, and a rare, high-pendency group. The study demonstrates both the power of careful feature engineering and the importance of honest baselines when evaluating ML models on small, structured public-policy datasets.

Index Terms—crime analytics, NCRB, India, regression, random forest, K-means clustering, time-series cross-validation, feature engineering, target leakage

I. INTRODUCTION

Crime statistics in India are published annually by the National Crime Records Bureau (NCRB) in the *Crime in India* series. Each annual report contains a disposal-of-cases table that tracks every IPC crime head through an “investigation pipeline”: cases pending from the previous year, cases reported during the year, cases reopened, cases disposed, and cases still pending. Two derived rates summarize the police response to each crime category:

- the *chargesheeting rate* — the fraction of disposed cases in which police filed a chargesheet, and
- the *pendency percentage* — the fraction of investigations unresolved at year end.

These two indicators are central to any quantitative assessment of policing effectiveness, yet no unified, cleaned, machine-readable version of the pipeline tables is publicly

available, and little published analysis attempts to *model* pipeline outcomes from upstream pipeline structure.

In this work we (i) harmonize the four NCRB disposal tables for 2018–2021, which differ substantially in column naming and layout; (ii) engineer a leakage-free feature set that captures caseload, backlog, and crime-category effects without algebraically encoding the target; (iii) benchmark seven predictive models against two principled baselines under time-ordered cross-validation; and (iv) apply unsupervised clustering to discover natural behaviour profiles among the 125 IPC crime heads. A key methodological finding is that a per-crime historical-mean baseline is very difficult to beat, which we interpret substantively rather than treat as a modelling failure.

II. RELATED WORK AND DATASET

The NCRB *Crime in India* reports are the canonical source of IPC crime statistics in India and are widely used in the policy and criminology literature. However, prior published use of these tables has been predominantly descriptive. The disposal-of-cases table in particular has received little supervised-learning treatment, in part because the column layout and naming change every year and a substantial data-engineering effort is required before any model can be fit across multiple years.

Each annual table reports, for every IPC crime head, columns for *cases pending from previous year*, *cases reported*, *cases reopened*, *total for investigation*, *cases not investigated*, *transferred*, *withdrawn*, five *final-report* subcolumns (non-cognizable, false, mistake of fact or law, insufficient evidence, abated), *chargesheets from previous year and during the year*, *total disposed*, *quashed and stayed*, and finally *cases pending at end of year*, from which the two summary rates are computed. We ingested the 2018, 2019, 2020, and 2021 tables and mapped each year’s column names onto a single canonical schema.

III. DATA HARMONIZATION AND FEATURE ENGINEERING

A. Schema Alignment

Column headers differed across years in three ways: (a) free-form punctuation and capitalization (for example, “Charge-Sheeting Rate” vs. “Charge Sheets Submitted”); (b) the pres-

ence of an explicit category column only in 2018; and (c) different Col. N references embedded in the header strings. We defined a dictionary-based renaming schema mapping the raw headers of each year onto 23 canonical fields. Crime-head names were then lowercased and whitespace-normalized, producing a field `crime_head_norm` that joins cleanly across years.

B. Row Filtering

The raw tables interleave category rows (e.g. “Total Offences against Property”), sub-total rows, and atomic crime-head rows. We built a set of *aggregate heads* from the category labels and excluded them. We then kept only those crime heads that appear in *all four* years to avoid introducing artificial year effects, yielding a master dataframe of 519 rows across 125 unique crime heads.

C. Category Attachment and Missing Values

Because only the 2018 table carries an explicit category column, we built a `CATEGORY_MAP` from the 2018 table and propagated it to 2019–2021 via `crime_head_norm`. Unmatched heads (a small tail) were labelled “Other IPC Crimes”. One row in 2018 had a missing `pendency_pct`; we imputed it with that year’s median.

D. Engineered Features

From the raw pipeline fields we derived:

- **Rates and Ratios**

$$\begin{aligned} \text{fr_rate} &= \frac{\text{fr_total}}{\text{total_disposed}} \\ \text{fr_false_rate} &= \frac{\text{fr_false}}{\text{total_disposed}} \\ \text{fr_insufficient_rate} &= \frac{\text{fr_insufficient}}{\text{total_disposed}} \\ \text{pendency_load} &= \frac{\text{cases_pending_prev}}{\text{total_for_inv}} \\ \text{reopened_rate} &= \frac{\text{cases_reopened}}{\text{cases_reported}} \\ \text{transferred_rate} &= \frac{\text{transferred}}{\text{total_for_inv}} \\ \text{cs_carryover_rate} &= \frac{\text{cs_from_prev}}{\text{cs_total}} \end{aligned}$$

- **Log-Transformed Features**

$$\begin{aligned} \log_cases_reported &= \log(1 + \text{cases_reported}) \\ \log_total_for_inv &= \log(1 + \text{total_for_inv}) \end{aligned}$$

- **Severity Ordinal** — a 1–5 ordinal constructed from the category (“Offences Affecting the Human Body” → 5; property-related → 2; documents and misc. → 1),
- a six-level one-hot encoding of the category, and
- `year_feature = year - 2018` to encode a monotonic temporal signal.

The cleaned dataframe was persisted as `pipeline_master.csv` for all downstream steps.

IV. EXPLORATORY DATA ANALYSIS

A. Target Distributions

Fig. 1 shows histograms of the two targets across all 519 rows. The chargesheeting rate has mean 73.5% and is strongly left-skewed with a mode near 85–90%, indicating that most IPC crime heads end in a chargesheet when disposed. The pendency percentage has mean 40.8% and is roughly symmetric with a right tail of chronically-backlogged crime types.

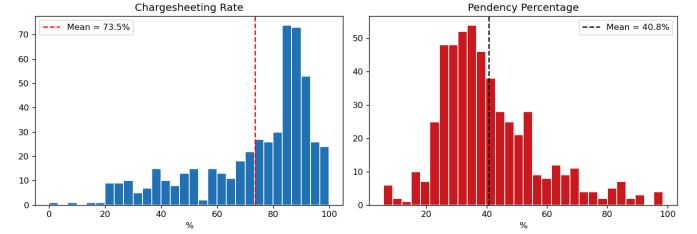


Fig. 1. Distributions of the two target variables over 519 crime-head-year observations, 2018–2021. Dashed lines show the means.

B. COVID-19 Effect on the Pipeline

Aggregating by year (Fig. 2), the total number of cases reported dipped in 2020 before rebounding in 2021, consistent with reduced reporting during lockdowns. More strikingly, *average pendency jumped in 2020* (from 40.1% to 44.1%) while the *chargesheeting rate stayed flat*, implying that courts and police were able to maintain disposal quality of what they did process, but inventory accumulated.

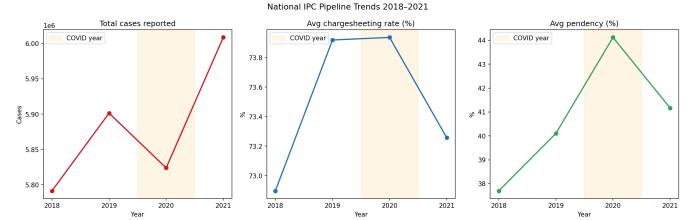


Fig. 2. National IPC pipeline trends 2018–2021. COVID-19 year (2020) shaded. Reported cases dipped while pendency spiked.

C. Correlation Screening and Target Leakage

Before fitting any model we computed the Pearson correlation matrix over all engineered numerical features including the two targets (Fig. 3). Two findings motivated our final feature selection:

- 1) `fr_rate` has correlation -0.99 with `chargesheeting_rate`, and `fr_insufficient_rate` has correlation -0.85 . These are algebraic near-inverses of the target (cases either end in a final report or a chargesheet), and including them would leak the target.
- 2) `pendency_load` has correlation $+0.88$ with `pendency_pct`. This is a meaningful (not algebraic)

dependency: backlog at year start structurally drives backlog at year end.

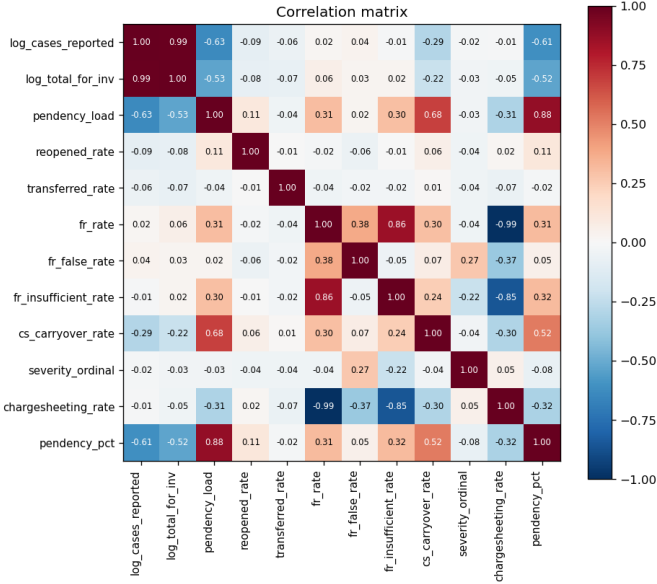


Fig. 3. Pearson correlation matrix of engineered features and targets. `fr_rate` and `fr_insufficient_rate` are algebraic near-inverses of `chargesheeting_rate` and were excluded from modelling.

Accordingly, we dropped `fr_rate`, `fr_false_rate`, and `fr_insufficient_rate` from the feature set but retained `pendency_load` (because it is causal rather than algebraic).

V. METHODOLOGY

A. Final Feature Set

The leakage-free feature set used for all supervised models comprises the five base features `{log_cases_reported, log_total_for_inv, pendency_load, severity_ordinal, year_feature}` together with the six category one-hot columns, giving 11 input features in total.

B. Time-Ordered Cross-Validation

Because the data are longitudinal, any uniform random split would leak future information into the past. We therefore use an expanding-window time-series cross-validation with three folds:

- fold 1: train on 2018, test on 2019;
- fold 2: train on 2018–2019, test on 2020;
- fold 3: train on 2018–2020, test on 2021.

For each model we report the mean of per-fold RMSE and MAE.

C. Baselines

We use two principled baselines rather than a single zero-rule:

- **Baseline 1 — Global Mean.** Predicts the overall train-set mean of the target for every test instance.

- **Baseline 2 — Per-Crime Mean.** Predicts the train-set mean of the target *within the same crime head*, falling back to the global mean for any unseen crime head.

Baseline 2 captures “this crime type behaves the same way every year” — a hypothesis the data strongly support and which any ML model must genuinely beat.

D. Models and Hyperparameter Search

We benchmarked five ML families. For each family we grid-searched over a small, principled set of hyperparameters:

- **Linear Regression** (on standardized inputs).
- **Ridge** with $\alpha \in \{0.01, 0.1, 1, 10\}$.
- **Lasso** with $\alpha \in \{0.01, 0.1, 1\}$, `max_iter` = 1500.
- **Random Forest** with `n_estimators` $\in \{100, 200\}$ and `max_depth` $\in \{5, 10, \text{None}\}$.
- **Support Vector Regression (RBF kernel)** with $C \in \{0.1, 1\}$, $\epsilon \in \{0.1, 0.5, 1\}$.
- **Gradient Boosting** with `n_estimators` $\in \{100, 200\}$, learning rate $\in \{0.01, 0.05, 0.1\}$.

Linear-family models (Linear, Ridge, Lasso) were trained on a `StandardScaler`-fitted design matrix, fit on each train fold and applied to the test fold to prevent scaler leakage.

E. Clustering

For the unsupervised analysis we collapsed the 519-row dataframe to a 125-row per-crime profile by averaging `{log_cases_reported, pendency_load, fr_rate, chargesheeting_rate, pendency_pct}` over years. After standardization we ran K-Means for $k \in \{2, \dots, 8\}$ and chose k by the peak silhouette score, cross-checked against the inertia elbow.

VI. RESULTS

A. Model Comparison

Table I reports cross-validated RMSE and MAE for every model, for both targets, with the best non-baseline model in each column highlighted in bold. Fig. 4 visualizes the same comparison against the per-crime-mean reference line.

TABLE I
TIME-SERIES CROSS-VALIDATION PERFORMANCE (MEAN OVER 3 FOLDS).
LOWER IS BETTER.

Model	chargesheeting_rate		pendency_pct	
	RMSE	MAE	RMSE	MAE
Baseline 1 (Global mean)	21.32	17.38	16.89	12.56
Baseline 2 (Per-crime mean)	<u>8.74</u>	<u>4.51</u>	8.52	5.43
Linear	17.72	13.63	8.36	5.61
Ridge ($\alpha = 0.01 / 10$)	17.72	13.62	8.31	5.59
Lasso ($\alpha = 0.01 / 1$)	17.72	13.63	8.03	5.33
Random Forest	15.21	10.22	8.13	5.35
SVR (RBF)	22.66	15.77	15.01	10.97
Gradient Boosting	15.80	11.23	8.07	5.68

Four observations follow:

- 1) **Per-crime mean is extremely strong for chargesheeting.** With RMSE = 8.74 it beats every supervised model,

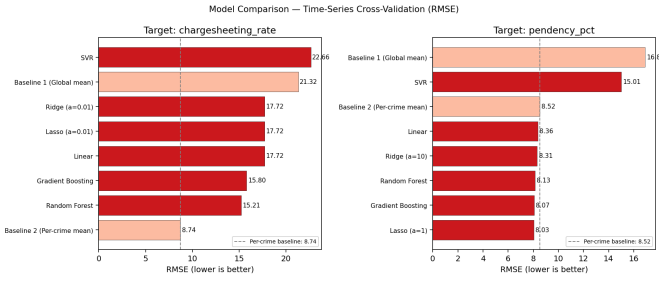


Fig. 4. Cross-validated RMSE for all models on both targets. The dashed vertical line is the per-crime-mean baseline. No ML model beats this baseline on chargesheeting rate; Lasso narrowly beats it on pendency percentage.

including Random Forest (15.21). This is a substantive finding: *year-to-year variation in a crime’s chargesheeting rate is small relative to differences between crime types*, and our pipeline features do not capture enough of the within-crime variation to improve on a “same crime, same behaviour” prior.

- 2) **Tree ensembles beat linear models** on chargesheeting rate by a large margin (RF 15.21 vs. Linear 17.72), consistent with non-linear interactions between caseload, backlog, and category.
- 3) **Pendency is easier to predict than chargesheeting.** For pendency, Lasso (RMSE 8.03) narrowly outperforms both tree ensembles and the per-crime baseline. Pendency is mechanically driven by backlog and caseload — both of which we observe directly — whereas chargesheeting depends on investigation quality, which we do not.
- 4) **SVR is not competitive** with default RBF settings on this data scale; it would benefit from more aggressive hyperparameter tuning that was out of scope here.

B. Feature Importance

We refit the best Random Forest configurations on all 519 rows and extracted Gini importances (Fig. 5):

- For `chargesheeting_rate`: `pendency_load` (0.283), `log_cases_reported` (0.232), `log_total_for_inv` (0.210), followed by the “Offences against Property” indicator (0.105).
- For `pendency_pct`: `pendency_load` dominates with importance 0.894, with all other features together contributing less than 0.11.

C. Clustering

Fig. 6 shows silhouette score and inertia as a function of k . Silhouette peaks sharply at $k = 3$ (0.459), and the inertia elbow lies between $k = 3$ and $k = 4$; we therefore use $k = 3$.

Table II reports the cluster centroids and representative crimes. The three clusters admit a clean operational interpretation:

- **Cluster 0 (73 crimes) — Standard, well-prosecuted.** Moderate caseload, low backlog, high chargesheeting (86%), low pendency (32%). Represents the bulk of IPC

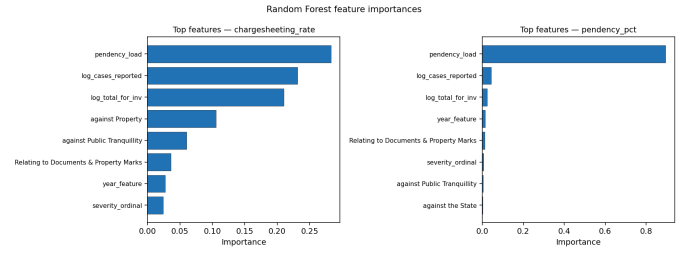


Fig. 5. Random Forest feature importances. `pendency_load` is the most informative single feature for both targets; it single-handedly drives pendency prediction.

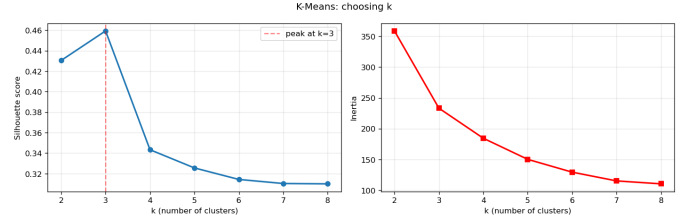


Fig. 6. Choosing k for K-Means. Silhouette peaks at $k = 3$ (left), confirmed by the inertia elbow (right).

crime heads (abetment of suicide, acid attack, affray, assault on women, attempt to commit culpable homicide, etc.).

- **Cluster 1 (23 crimes) — Property-crime backlog.** Highest caseload, moderate backlog, very low chargesheeting (40%) and high final-report rate (60%). Contains arson, ATM/bank theft, auto theft, burglary, credit-card fraud, criminal misappropriation. These are crimes with routinely weak evidence trails, ending disproportionately in final reports.
- **Cluster 2 (29 crimes) — Rare, high-pendency.** Small caseload, highest backlog (pendency 62%), moderate chargesheeting. Contains counterfeiting offences, cheating by impersonation, communal/religious offences, trafficking-related offences. Low volume combined with difficult investigation profiles.

TABLE II
K-MEANS ($k = 3$) CLUSTER CENTROIDS OVER THE FIVE PROFILE FEATURES.

Cluster (n)	log_cases reported	pend. load	fr_rate	CS rate (%)	pend. (%)
0 (73)	9.05	0.31	0.14	86.11	32.30
1 (23)	9.53	0.41	0.60	39.64	42.45
2 (29)	4.93	0.61	0.30	69.76	61.80

The fact that an unsupervised method, with no access to category labels, recovers three groups that align with property-crime-versus-other is a useful cross-check on the supervised feature importances.

VII. DISCUSSION

A. Why the Per-Crime Baseline Is So Strong

The result that no supervised model beats Baseline 2 on `chargesheeting_rate` deserves emphasis, not embarrassment. With 125 crime heads and only four years of data, each crime contributes on average 4.15 observations. Within any single crime, the chargesheeting rate is dominated by the nature of the offence (how much physical or documentary evidence tends to exist), which is stable over a 4-year horizon. The remaining year-to-year variance is small. Our pipeline features, which describe caseload and backlog rather than evidentiary quality, are essentially orthogonal to this residual variance. This is an important negative result: it suggests that improving chargesheeting-rate prediction in a future study would require features describing *investigation quality* — for example, per-district officer strength, forensic turnaround times, or digitization levels — rather than more pipeline columns.

B. The Target-Leakage Lesson

Our initial feature set contained `fr_rate`, `fr_false_rate`, and `fr_insufficient_rate`. Because cases are disposed as either a chargesheet or a final report, and the chargesheeting rate is the complement of the final-report rate up to small remainders, these features produced near-perfect apparent performance ($\text{RMSE} \ll 1$) that collapsed as soon as they were removed. We emphasize this as a reproducible cautionary tale: *in any dataset derived from the same pipeline as the target, a correlation check at $|r| > 0.8$ should precede model fitting*, and any feature that is an algebraic near-identity of the target must be dropped.

C. Clustering as Interpretive Scaffolding

The three-cluster solution is actionable in a way the supervised models alone are not. A policy maker can point to Cluster 1 — property-volume crimes with a chronic chargesheeting deficit — and ask targeted questions about evidentiary infrastructure (CCTV, serial-number registries, forensic throughput), separately from the investigation-capacity questions raised by Cluster 2 (rare, specialized crimes with chronic backlog).

D. Limitations

Three limitations shape the scope of the conclusions:

- **Only four years of data.** 519 rows is tight for ML; more years (or monthly granularity) would tighten variance estimates substantially.
- **National aggregates only.** The NCRB tables we used are national totals. State-level disaggregation would almost certainly unlock richer state-by-category interactions.
- **No investigation-quality features.** As argued above, the chargesheeting-rate ceiling set by Baseline 2 is unlikely to move without features that describe how an investigation was conducted, not merely what flowed through the pipeline.

VIII. CONCLUSION

We built the first, to our knowledge, cleaned multi-year NCRB pipeline dataset and benchmarked a family of supervised and unsupervised methods against honest time-ordered baselines. Random Forest was the strongest supervised learner for chargesheeting rate (RMSE 15.21) but could not match the per-crime historical mean (RMSE 8.74), a finding that pushes future work toward investigation-quality features rather than more pipeline engineering. For pendency percentage, Lasso was marginally best at RMSE 8.03, and a single engineered feature — `pendency_load` — explained 89% of the Random Forest’s predictive power. K-Means on the per-crime profile cleanly recovered three interpretable operational groups that align with property-versus-human-body-versus-rare-specialized crime categories. The study illustrates how careful feature engineering and principled baselines together yield an honest, actionable picture of a public-policy dataset, and identifies the specific data-collection gap that next-generation work on Indian crime analytics should target.

ACKNOWLEDGMENT

The authors thank the National Crime Records Bureau (NCRB), Ministry of Home Affairs, Government of India, for making the *Crime in India* reports publicly available.

The authors also express their sincere gratitude to Prof. Amit Sethi, Department of Electrical Engineering, Medical Deep Learning and Artificial Intelligence Lab (MeDAL), CMiNDS & KCDH, IIT Bombay, for his guidance and support through the course EE 769: Introduction to Machine Learning.

REFERENCES

- [1] National Crime Records Bureau, “Crime in India 2018,” Ministry of Home Affairs, Government of India, 2019.
- [2] National Crime Records Bureau, “Crime in India 2019,” Ministry of Home Affairs, Government of India, 2020.
- [3] National Crime Records Bureau, “Crime in India 2020,” Ministry of Home Affairs, Government of India, 2021.
- [4] National Crime Records Bureau, “Crime in India 2021,” Ministry of Home Affairs, Government of India, 2022.
- [5] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [6] R. Tibshirani, “Regression shrinkage and selection via the lasso,” *Journal of the Royal Statistical Society: Series B*, vol. 58, no. 1, pp. 267–288, 1996.
- [7] J. H. Friedman, “Greedy function approximation: a gradient boosting machine,” *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [8] P. J. Rousseeuw, “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis,” *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.
- [9] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.